

05.1

Pears are a type of fruit.

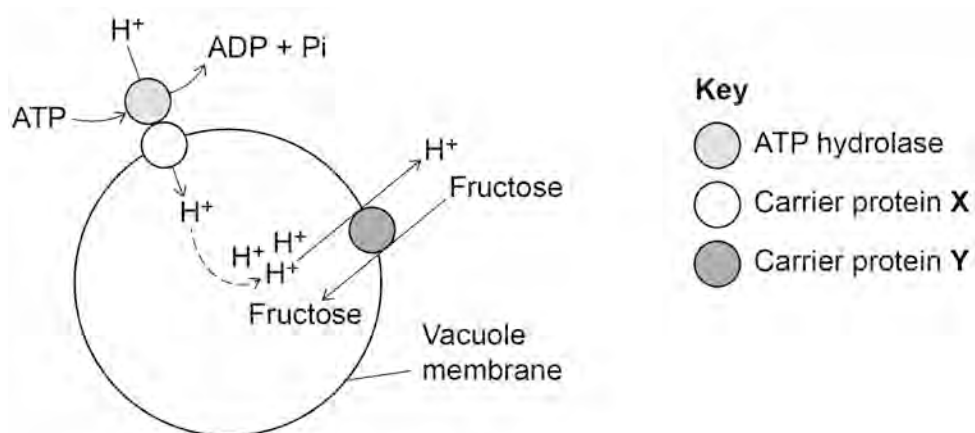
When pears ripen, the:

- fructose concentration increases in cell vacuoles
- volume of cell vacuoles increases.

Figure 2 shows information about a pear cell's vacuole membrane.

- Carrier protein **X** is linked to an ATP hydrolase enzyme.
- Carrier protein **Y** co-transporters H^+ ions and fructose molecules in opposite directions.

Figure 2



Explain why the volume of the vacuoles in pear cells increases when pears ripen.

Your answer should refer to the role of:

- ATP hydrolase
- carrier protein **X** and carrier protein **Y**.

[4 marks]

0	5	.	2
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A respiratory inhibitor caused a change in pH of the solution within the vacuole.

Use information in **Figure 2** to explain how.

[3 marks]

0	5	.	3
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Explain how a change in the expression of the genes coding for the carrier proteins can increase the rate of ripening of pears.

[2 marks]
